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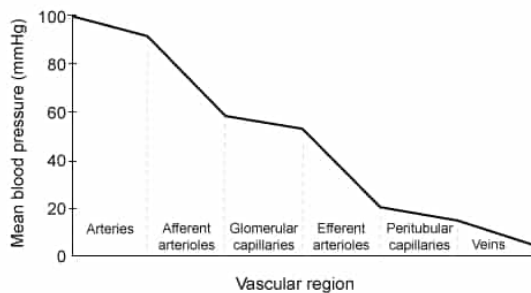


Figure 1 Mean blood pressure in different regions of the kidney vasculature

Pressure-sensitive nerves in the renal arteries that supply blood to the kidneys are important for regulating blood pressure. Unfortunately, these nerves can malfunction and cause high blood pressure, which is associated with several health risks including strokes, heart attacks, and hardening of the arteries (ie, arteriosclerosis). When arteries harden, the blood pressure spike generated from each heartbeat, termed the pulse wave, travels faster through the vasculature and can physically damage the vessel walls.

Renal artery ablation destroys the malfunctioning nerves and normalizes blood pressure. Ablation transmits energy, such as radio frequency (RF) or laser energy, to heat the tissue. Researchers tested an RF ablation device and determined it delivered 100 J of thermal energy into 3 g of tissue.

Researchers also tested a laser ablation system consisting of a glass optical fiber coupled to various lenses for focusing the light. The incident angle of the laser in the glass fiber was held constant at 45° . The refracted angles for a polymer lens and a liquid crystal lens were 55° and 35° , respectively.

Researchers determined that the blood velocity in the afferent arterioles of their model was four times larger than physiological measurements. Which of the following changes to their model could correct this discrepancy? (Note: Assume ideal fluid flow.)

- ✔ ☒ A. Increase the radius of the afferent arterioles by 100% [37%]
- ☐ B. Increase the radius of the afferent arterioles by 75% [16%]
- ☐ C. Decrease the radius of the afferent arterioles by 75% [26%]
- ☐ D. Decrease the radius of the afferent arterioles by 100% [20%]

Correct

36% answered correctly

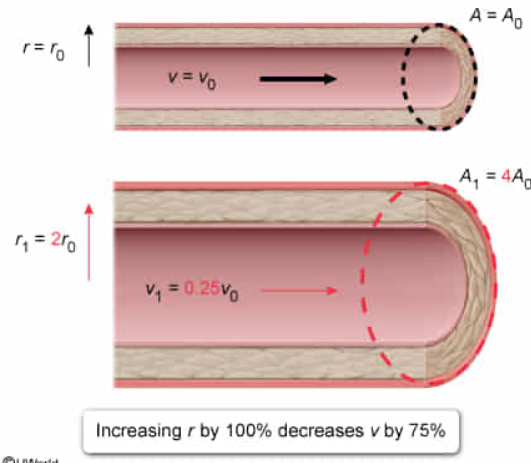
Explanation:

Effect of blood vessel radius on blood velocity

$$Q = Av$$

$$v = \frac{Q}{A} = \frac{Q}{\pi r^2}$$

- Q Volumetric flow rate
- A Area of vessel
- v Blood velocity
- r Radius of vessel



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The **volumetric flow rate** Q of blood flowing through a vessel equals the product of the cross-sectional **area** A of the blood vessel and the **velocity** v of the blood:

$$Q = Av$$

Furthermore, A is equal to the product of π and the square of the vessel's **radius** r . Substituting this relationship into the equation above yields:

$$Q = \pi r^2 v$$

In this question, Q is constant over all the vascular regions in the kidney vasculature model. The equation above can be rearranged to solve for v :

$$v = \frac{Q}{\pi r^2}$$

Hence, v is **inversely proportional** to r^2 .

The value of v for the original vessel r is denoted v_0 and equals:

$$v_0 = \frac{Q}{\pi r^2}$$

When r increases by 100% (ie, is doubled), the new value of v , denoted v_1 , equals:

$$v_1 = \frac{Q}{\pi(2r)^2} = \frac{Q}{4\pi r^2}$$

$$v_1 = 0.25 \frac{Q}{\pi r^2} = 0.25v_0$$

Therefore, increasing r by 100% causes v to decrease by 75% (ie, decrease to one fourth its original value), in agreement with the **continuity equation**.

(Choice B) Increasing the radius of the arterioles by 75% would decrease the blood velocity by 68% (ie, decrease to approximately one third its original value), not one fourth its original

value.

(Choices C and D) Decreasing the radius of the arterioles would cause the blood velocity to increase, not decrease.

Educational objective:

The volumetric flow rate of blood is constant in all the vascular regions of the kidney. The blood velocity equals the ratio of the volumetric flow rate and the blood vessel cross-sectional area.

Time spent:0 sec

Subject:
Physics

QID:403631

System:
Fluids

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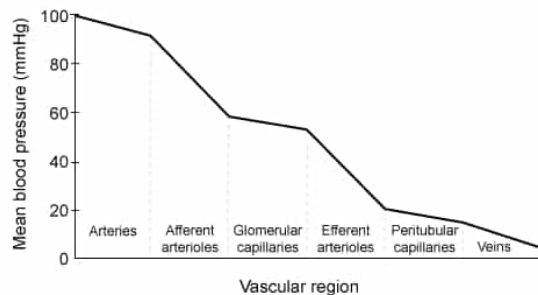


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What is the temperature change of the renal artery from the RF ablation system described in the passage? (Note: The specific heat of renal artery tissue is $3.3 \text{ J}/(\text{g}\cdot^\circ\text{C})$.)

- ☐ A. 1°C [3%]
- ☐ B. 3°C [2%]
- ☒ C. 10°C [89%]
- ☐ D. 30°C [4%]

Correct

88% answered correctly

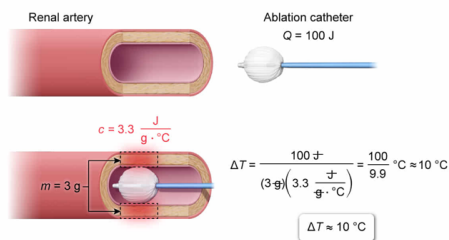
Explanation:

Temperature change during renal artery ablation

$$Q = mc\Delta T$$

$$\Delta T = \frac{Q}{mc}$$

- Q Heat
- m Mass of tissue
- c Specific heat
- ΔT Temperature change



The **calorimetry** equation states that the **heat** Q transferred into an object equals the product of the object's **mass** m , its **specific heat** c , and the change in **temperature** ΔT :

$$Q = mc \Delta T$$

In this question, the value of ΔT is calculated by rearranging the above equation:

$$\Delta T = \frac{Q}{mc}$$

The Q transferred into the renal artery is 100 J, the m of the artery is 3 g, and the c is 3.3 J/(g·°C). Substituting these values into the equation above yields:

$$\Delta T = \frac{100 \text{ J}}{(3 \text{ g}) \left(3.3 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}} \right)} = \frac{100}{9.9} ^\circ\text{C} \approx 10 ^\circ\text{C}$$

Therefore, the renal artery ablation system raises the tissue temperature by 10 °C.

(Choice A) 1 °C results from calculating the ratio of the specific heat and the mass of the artery.

(Choice B) 3 °C is calculated by using 10 J for heat and omitting the mass of the artery.

(Choice D) 30 °C is obtained by omitting the mass of the artery from the calculation.

Educational objective:

The heat energy required to raise the temperature of an object depends on the object's mass and specific heat. The change in the object's temperature equals the ratio of the heat and the product of the mass and specific heat.

Time spent: 0 sec

Subject:
Physics

QID: 403632

System:
Fluids

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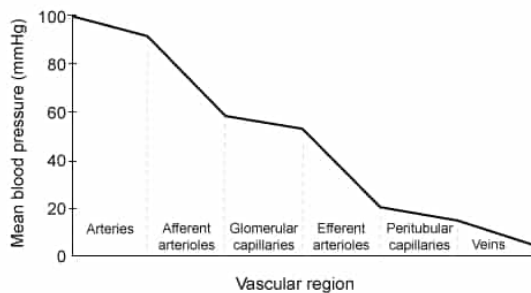


Figure 1 Mean blood pressure in different regions of the kidney vasculature

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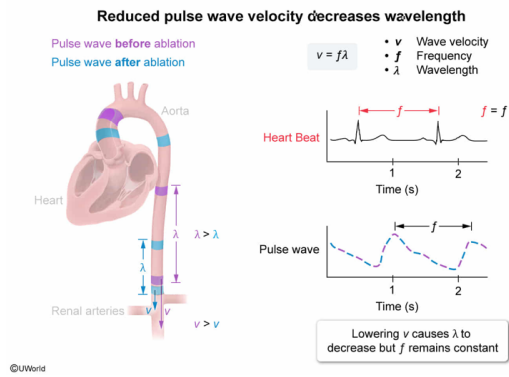
A clinical study reports that the time required for the pulse wave to arrive at the kidney increases after renal artery ablation. A possible explanation for this could be that:

- ☒ A. decreasing blood pressure causes the wavelength of the pulse wave to decrease. [15%]
- ☐ B. decreasing blood pressure causes the wavelength of the pulse wave to increase. [17%]
- ☐ C. decreasing blood pressure causes the frequency of the pulse wave to decrease. [53%]
- ☐ D. decreasing blood pressure causes the frequency of the pulse wave to increase. [12%]

Correct

14% answered correctly

Explanation:



For each heartbeat, the **pulse wave** travels through the vasculature, reaching different arteries at different times depending on their distance from the heart. The **frequency** f of the pulse wave is equal to the heart rate. The **wave velocity** v depends on vascular parameters, such as the vessel stiffness. The **wavelength** λ is the distance between the peaks of the pulse wave. The value of v equals the product of f and λ :

$$v = f\lambda$$

In this question, ablation of the renal arteries causes an increase in the travel time for the pulse wave to reach the kidney. Therefore, the pulse wave v *decreases* after renal artery ablation. Because f depends only on the heart rate, it does not change before and after ablation. Furthermore, according to the equation above, λ is **directly proportional** to v . Therefore, λ must decrease when v is decreased.

(Choice B) A decrease in pulse wave velocity causes the wavelength to decrease, not increase.

(Choices C and D) The pulse wave frequency depends only on the heart rate and is the same before and after ablation.

Educational objective:

The pulse wave velocity is equal to the product of the frequency and the wavelength. When the frequency is constant, changes in the pulse wave velocity cause a proportional change in the wavelength.

Time spent: 0 sec

Subject:
Physics

QID: 403633

System:
Fluids

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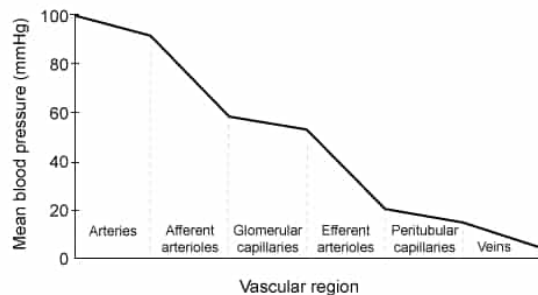


Figure 1 Mean blood pressure in different regions of the kidney vasculature

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If the maximum pressure in a blood vessel is 7,000 Pa, what is the pressure in mmHg?

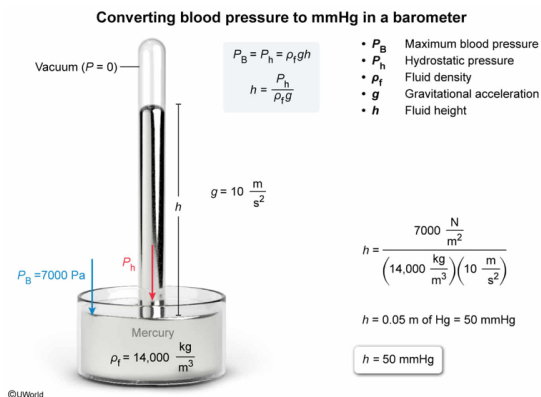
(Note: The density of mercury is $14,000 \text{ kg/m}^3$, and gravitational acceleration is 10 m/s^2 .)

- ☐ A. 0.05 mmHg [7%]
- ☐ B. 2 mmHg [9%]
- ☐ C. 20 mmHg [27%]
- ☒ D. 50 mmHg [55%]

Correct

56% answered correctly

Explanation:



Pressure refers to a force exerted over an area, such as the atmospheric pressure P_{atm} caused by Earth's atmosphere pushing in on any object. P_{atm} can be measured from the height of a column of mercury in a **barometer**. The P_{atm} of the air pushing downward equals the **hydrostatic pressure** P_h of the mercury column, which equals the product of the **fluid density** ρ_f , **gravitational acceleration** g , and the **fluid height** h :

$$P_h = \rho_f g h = P_{\text{atm}}$$

The equation above can be solved for h , yielding:

$$h = \frac{P_h}{\rho_f g}$$

In this question, P_h in the mercury barometer equals the maximum blood vessel pressure P_B , which is given as 7,000 **Pa**. From the equation above, h in meters of mercury is given by:

$$h = \frac{7,000 \frac{\text{N}}{\text{m}^2}}{\left(14,000 \frac{\text{kg}}{\text{m}^3}\right) \left(10 \frac{\text{m}}{\text{s}^2}\right)}$$

$$h = \frac{7}{140} \left(\frac{\text{kg} \cdot \text{m}}{\text{s}^2 \cdot \text{m}^2} \right) \left(\frac{\text{m}^3}{\text{kg}} \right) \left(\frac{\text{s}^2}{\text{m}} \right) = 0.05 \text{ m of Hg}$$

Converting this value to **mmHg** yields:

$$h = 0.05 \text{ m of Hg} \left(\frac{1,000 \text{ mmHg}}{1 \text{ m of Hg}} \right) = 50 \text{ mmHg}$$

(Choice A) 0.05 mmHg is obtained by not converting the units from meters of mercury into mmHg.

(Choice B) 2 mmHg results from calculating the ratio of the fluid density and the hydrostatic pressure instead of the ratio of hydrostatic pressure and the product of fluid density and gravitational acceleration.

(Choice C) 20 mmHg is calculated from the product of gravitational acceleration and the ratio of fluid density and hydrostatic pressure instead of the ratio of hydrostatic pressure and the product of fluid density and gravitational acceleration.

Educational objective:

The hydrostatic pressure in a mercury barometer equals the product of the fluid density, gravitational acceleration, and the height of the mercury in the tube.

Time spent:0 sec

Subject:

Physics

QID:403634

System:

Fluids

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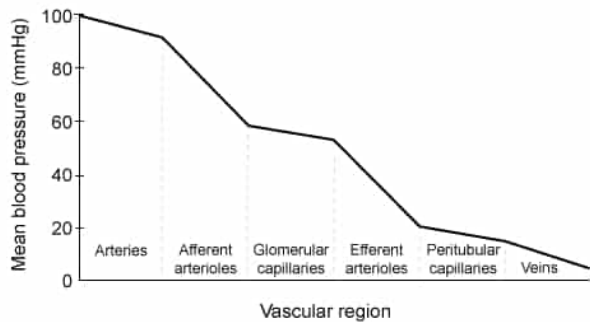


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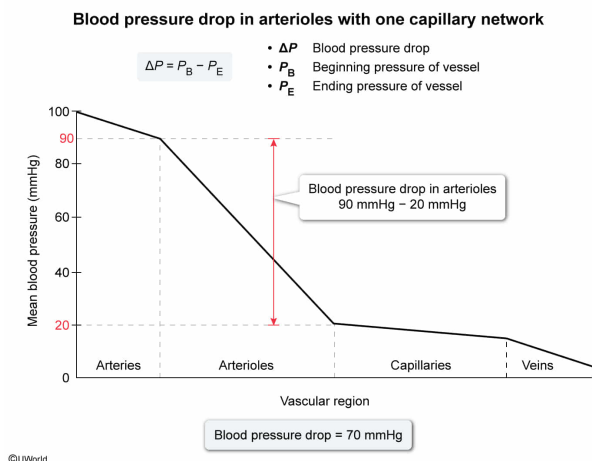
Which of the following is closest to the expected drop in pressure in the arterioles for an organ with only one capillary network?

- ☐ A. 100 mmHg [4%]
- ✓ ☒ B. 70 mmHg [27%]
- ☐ C. 40 mmHg [45%]
- ☐ D. 20 mmHg [22%]

Correct

27% answered correctly

Explanation:



Blood circulates through different **vascular regions** as it perfuses the bodily organs. Blood is pumped out of the heart and typically traverses the following vascular regions: arteries, **arterioles**, **capillaries**, and finally veins. Each vascular region has a characteristic **vascular resistance** that contributes to the overall drop in **blood pressure** across the circulatory system.

In this question, the mean blood pressure in different vascular regions of the kidney is provided in Figure 1. However, the kidney has an unusual vascular architecture with two capillary networks. In an organ with only one capillary network, the arterioles must reduce the blood pressure at the end of the arteries, approximately 90 mmHg, to the value at the beginning of the second capillary network, approximately 20 mmHg. Therefore, the pressure drop across the arterioles is 70 mmHg.

(Choice A) 100 mmHg is the total blood pressure drop across all of the vascular regions, not the pressure drop across the arterioles.

(Choice C) 40 mmHg is the blood pressure drop across the glomerular capillaries and the efferent arterioles in the kidney.

(Choice D) 20 mmHg is the blood pressure at the beginning of the capillary network.

Educational objective:

The arterioles are responsible for reducing the blood pressure from the major arteries (high blood pressure) to the capillary networks (low blood pressure). As a result, the arterioles are the primary pressure-reducing vascular region of the circulatory system.

Time spent: 0 sec
Subject:
Physics

QID: 403635
System:
Fluids

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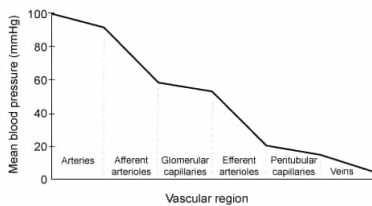


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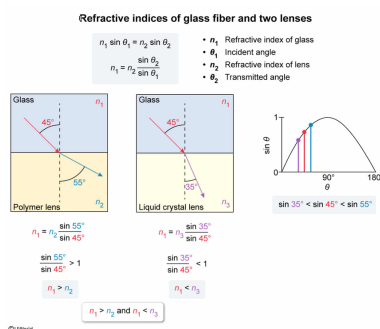
The refractive indices of the glass optical fiber, polymer lens, and liquid crystal lens described in the passage are denoted as n_1 , n_2 , and n_3 , respectively. The refractive indices of the glass optical fiber and the two focusing lenses are related by which of the following?

- ☐ A. $n_1 < n_2$ and $n_1 < n_3$ [6%]
- ☐ B. $n_1 > n_2$ and $n_1 > n_3$ [7%]
- ☐ C. $n_1 < n_2$ and $n_1 > n_3$ [40%]
- ☒ D. $n_1 > n_2$ and $n_1 < n_3$ [45%]

Correct

46% answered correctly

Explanation:



When **light** encounters a boundary between two materials with different **refractive indices**, n_1 and n_2 , it **refracts**, causing its path to bend as it crosses the boundary. As a result, the **incident angle** θ_1 of the light is different than the **refracted angle** θ_2 , with both angles measured relative to a line perpendicular to the boundary. The relationship between these angles and the refractive indices are given by **Snell's law**:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

This equation can be solved for n_1 , yielding:

$$n_1 = n_2 \frac{\sin \theta_2}{\sin \theta_1}$$

In this question, the relationships between n_1 and n_2 and between n_1 and n_3 are given by the equation above:

$$n_1 = n_2 \frac{\sin \theta_2}{\sin \theta_1} = n_2 \frac{\sin 55^\circ}{\sin 45^\circ}$$

$$n_1 = n_3 \frac{\sin \theta_3}{\sin \theta_1} = n_3 \frac{\sin 35^\circ}{\sin 45^\circ}$$

The **sine function** monotonically increases as the angle increases from 0° to 90° . Therefore, the ratios in the equations above are:

$$\frac{\sin 55^\circ}{\sin 45^\circ} > 1$$

$$\frac{\sin 35^\circ}{\sin 45^\circ} < 1$$

Therefore, n_1 equals the product of n_2 and a value greater than 1, forcing n_1 to be greater than n_2 :

$$n_1 > n_2$$

Similarly, n_1 equals the product of n_3 and a value less than 1, requiring n_1 to be less than n_3 :

$$n_1 < n_3$$

(Choice A) If n_1 were less than n_2 , θ_2 would be less than θ_1 , but θ_2 is 55° and θ_1 is 45° .

(Choice B) If n_1 were greater than n_3 , θ_3 would be greater than θ_1 , but θ_3 is 35° and θ_1 is 45° .

(Choice C) The ratio of the refractive indices $\frac{n_1}{n_2}$ is related to the ratio of the angles, $\frac{\theta_2}{\theta_1}$. Therefore, n_1 is greater than n_2 and n_1 is less than n_3 , because θ_1 is less than θ_2 and θ_1 is greater than θ_3 .

Educational objective:

Snell's law governs the refraction of light when crossing a boundary of two media with different refractive indices. Light traveling from a medium with a higher refractive index into one with a lower refractive index will have a refracted angle that is greater than the incident angle.

Time spent: 0 sec

Subject:
Physics

QID: 403636

System:
Fluids